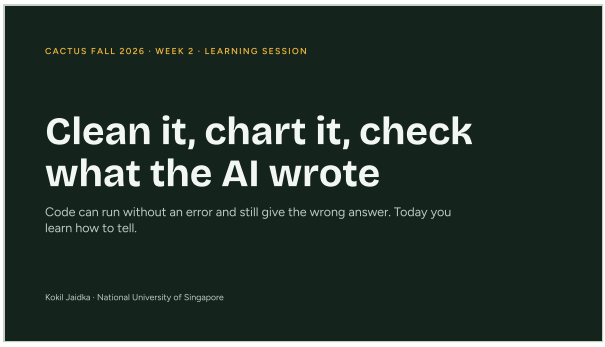
# CACTUS Week 2: Clean It, Chart It



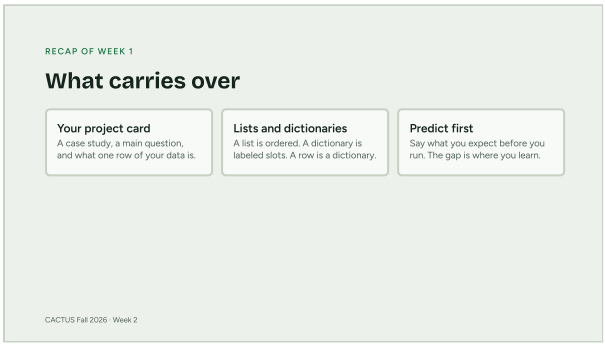

## Slide 1

**TIME** 2 min (0:00)

**SAY** Welcome back. This week is about AI as pair programmer: you read it, test it, and do not trust it blindly. By the end of the week you can clean a table, say what each step did to the row count, and prove a suspected bug with a check.

**DO** Welcome people by name where you can. Post the course site link. Ask who did the practice page or reading (hands or chat), without making anyone feel bad if not.

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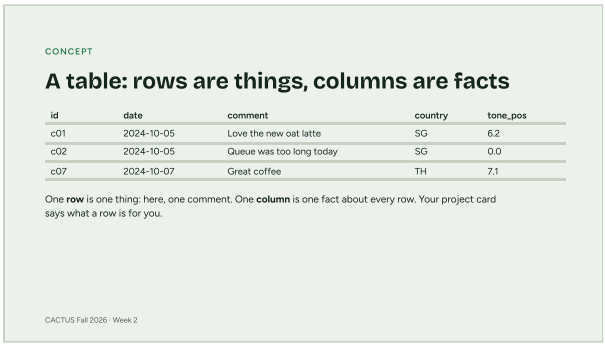

## Slide 2

**TIME** 3 min

**SAY** Quick recap. Your project card. Lists and dictionaries: a row of data is a dictionary. And the habit: predict before you run.

**DO** Ask one or two people to say their case study in one sentence. Keep it short. This reminds everyone what the week should serve.

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## Slide 3

**TIME** 6 min

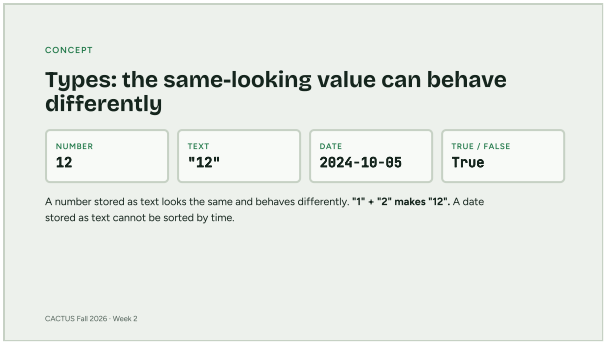
**SAY** A table: rows are things, columns are facts about them. Here one row is one comment. Your project card says what a row is for you.

**DO** Ask the room: what is one row in your project? Take two or three answers. Point out that a pandas DataFrame is the same idea as a spreadsheet.

**IF ASKED "What is pandas?"** A Python tool for working with tables. Chapter 5 of the textbook introduces it.

**WATCH FOR** No syntax yet. Keep it conceptual.

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## Slide 4

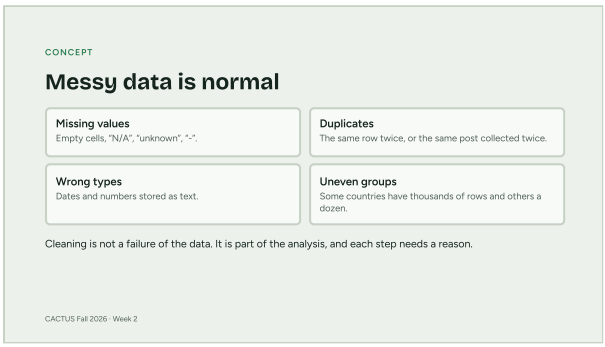
**TIME** 5 min

**SAY** The same-looking value can behave differently. 12 as a number, "12" as text, a date, true or false. Remember the puzzle page: quotes made a value text.

**DO** Point at 12 and "12". Say: add one and one as text and you get 11, not 2. Then say: in real data this happens silently when one bad cell, like "unknown", turns a whole column into text.

**IF ASKED how to check** df.info() shows each column's type. We will use it in the notebook.

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## Slide 5

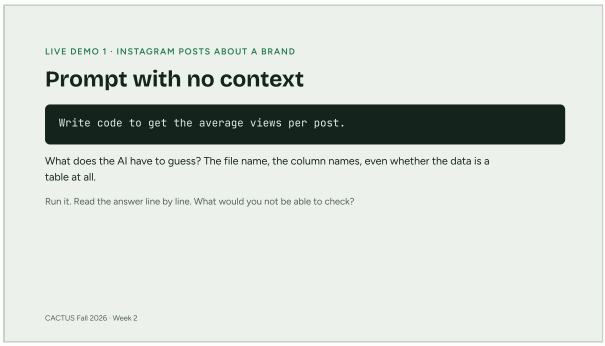
**TIME** 4 min (ends 0:20)

**SAY** Messy data is normal. Missing values, duplicates, wrong types, uneven groups. Cleaning is part of the analysis, and each step needs a reason you can say out loud.

**DO** Read the four cards. Say: you will see all four today and in the notebook.

**WATCH FOR** People who feel the data is wrong. Say: it is not wrong, it is real.

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## Slide 6

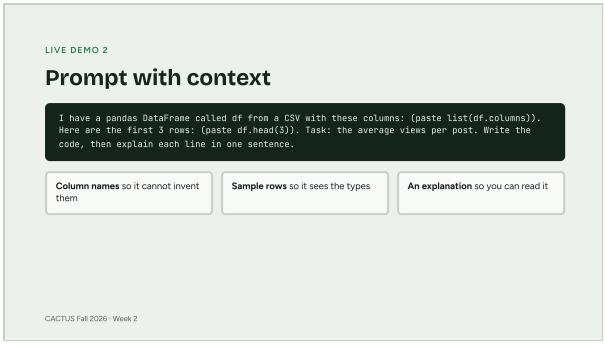
**TIME** 18 min (0:20 to 0:38)

**DO (live demo)** Open the Lululemon file (lululemon_instagram.csv) in Colab or a notebook, share your screen, and open an AI chat. Paste ONLY the prompt on the slide: "Write code to get the average views per post." Do not mention the file, columns, or any context.

**SAY** Watch what the AI has to guess: the file name, the column names, even whether this is a table.

**DO** Read its answer line by line with the room. Ask: which names did it make up? What would you not be able to check? Run it if you like and show the error or odd result.

**WATCH FOR** Do not skip to the good prompt. Let the failure be uncomfortable for a minute. That is the lesson.




## Slide 7

**TIME** 22 min (0:38 to 1:00)

**DO (live demo)** In the notebook, run df.columns and df.head(3) on the Lululemon data. Copy the output into the prompt on the slide, then submit it. Compare the answer to the first attempt.

**SAY** Context removed the guessing. Column names so it cannot invent them. Sample rows so it sees the types. An explanation so you can read it.

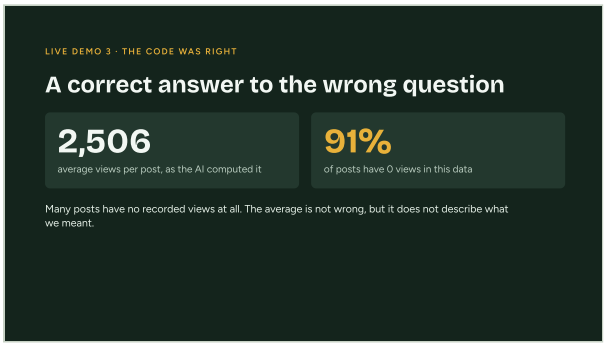
**DO** Read the new code line by line. For each line ask: what does this do to my data? Run it.

**SAY (reviewer's rule)** If you cannot explain a line, reject it.

**WATCH FOR** Never paste private data into an AI chat. This dataset is anonymized, but say it out loud as a habit.

**NEXT** Do not reveal the 2,506 result yet if you want to ask the room to predict it. Move straight to the next slide.

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## Slide 8

**TIME** 12 min (1:00 to 1:12)

**DO** Run the AI's average views. It prints about 2,506. Then ask the room: how many posts have 0 views? Run (df["Views"] == 0).mean(). It shows about 91 percent.

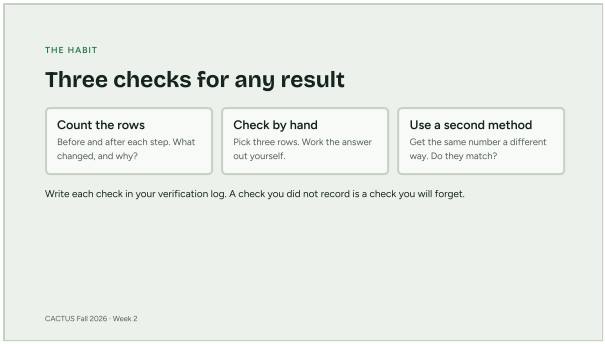
**SAY** The code is correct. The question was wrong. In this data a zero often means not recorded, not that nobody viewed it. An average over mostly zeros does not describe what we meant.

**SAY** Running without error is not the same as being right. That is the lesson of the week.

**IF ASKED how to handle it** In Week 2 you will see options. For now the point is to notice.

**WATCH FOR** Do not make the AI the villain. The AI did what we asked. We did not check the data first.

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## Slide 9

**TIME** 8 min (1:12 to 1:20)

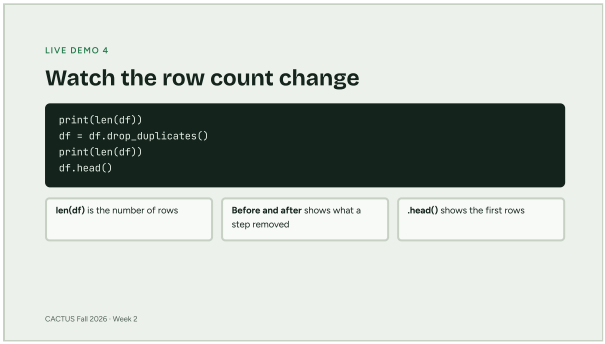
**SAY** Three checks for any result. Count the rows before and after each step. Check a few rows by hand. Get the same number a second way.

**DO** For each, give a one-line example from the Lululemon demo: row counts (len before and after), by hand (pick three posts and look at their views), second method (compare mean with the median or the count of zeros).

**SAY** Write each check in your verification log. A check you did not record is a check you will forget.

**WATCH FOR** People who think they have to do all three every time. Say: choose the cheapest one that could catch the mistake you are worried about.

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## Slide 10

**TIME** 10 min (1:20 to 1:30)

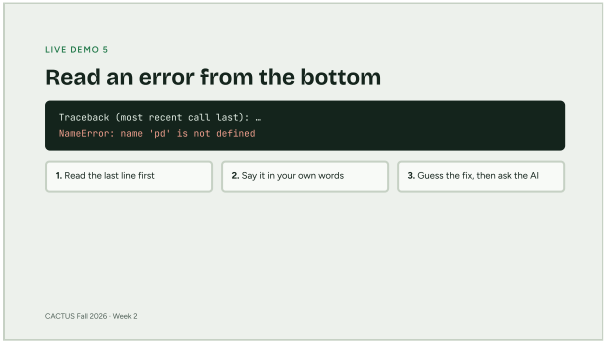
**DO (live demo)** In the notebook run print(len(df)), then df = df.drop_duplicates(), then print(len(df)). Before the second print, ask the room to predict the number. Then run df.head().

**SAY** len is the number of rows. Before and after shows what a step removed. head shows the first few rows. This is the most useful ten seconds of debugging you will learn this week.

**IF THE COUNT DOES NOT CHANGE** say, good, that tells us there were no exact duplicates, and move on.

**WATCH FOR** People who want to type along. Let them, but keep the demo moving.

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## Slide 11

**TIME** 20 min (1:30 to 1:50)

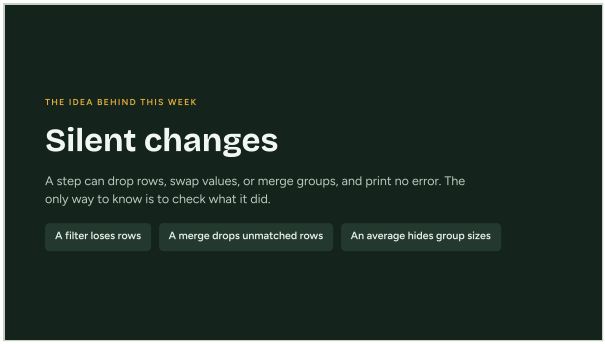
**DO (live demo)** In a fresh notebook, run a cell that uses pd before you run the import cell. You get: NameError: name 'pd' is not defined. Do not tell the room what it means yet.

**SAY** Three steps. Read the last line first. Say it in your own words. Guess the fix, then ask the AI.

**DO** Ask for volunteers to say the last line in their own words (it says Python does not know the name pd yet). Then paste the full message into an AI and compare its explanation to the volunteers'. Apply the fix and say in one sentence why it works.

**DO (second, if time)** Run a cell with a deliberate typo in a column name and repeat the three steps.

**WATCH FOR** People panicking at red text. Say: errors are information, and the last line is the most useful part.




## Slide 12

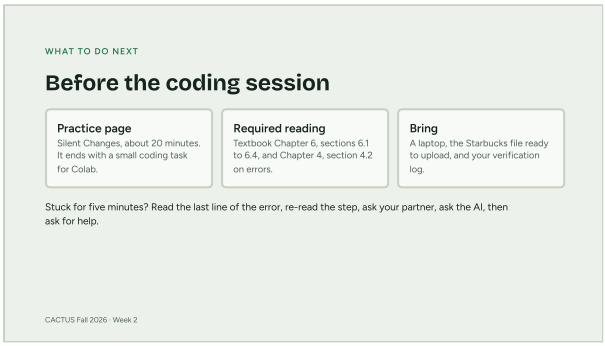
**TIME** 4 min (1:50)

**SAY** This is the idea behind the practice page. A step can drop rows, swap values, or merge groups and print no error. The only way to know is to check what it did.

**DO** Name the three examples on the slide. Say: the practice page has you build a cleaning pipeline and watch the row count change, then hunt three bugs that run without errors.

**NOTE** The practice page uses bug types that are different from the live exercise, so it does not give that away. Do not describe the live exercise bugs here.

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## Slide 13

**TIME** 6 min (1:54 to 2:00)

**SAY** Before the coding session: the practice page (about 20 minutes), the required reading (textbook Chapter 6 sections 6.1 to 6.4 and Chapter 4 section 4.2), and bring your laptop and the Starbucks file ready to upload.

**DO** Post the practice page and the course site link in the chat. Read out the stuck protocol: five minutes, read the last line, re-read the step, ask your partner, ask the AI, then ask for help.

**SAY** In the coding session there is no lecture. You work in pairs in breakout rooms and facilitators visit.

**CLOSE** Homework is to finish Exercise 5 on your own data and add one line to your project card on what could go wrong in your data. Thank everyone.